Jiawei Fu

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Google Scholar | LinkedIn | Github

Education

University of California San Diego, Ph.D. advised by **Prof. Henrik Christensen** Sept. 2024 – PRESENT

- Develop **data collection systems for diverse robots** and **efficient learning methods for robotic manipulation**

ETH Zurich, M.Sc. in Robotics, Systems and Control Sept. 2020 – Feb. 2024

- GPA: 5.76/6

Tsinghua University, B.Eng. in Mechanical Engineering (Elite Program) Sept. 2016 – Jun. 2020

- GPA: 3.82/4 (**ranked 1st in the elite program**)

Publications

LodeStar: Long-horizon Dexterity via Synthetic Data Augmentation from Human Demonstrations **CORL 2025**

Weikang Wan*, **Jiawei Fu***, Xiaodi Yuan, Yifeng Zhu, Hao Su

How Far Can a 1-Pixel Camera Go? Solving Vision Tasks Using Photoreceptors and Computationally Designed Visual Morphology **ECCV 2024**

Andrei Atanov*, **Jiawei Fu***, Rishubh Singh*, Isabella Yu, Andrew Spielberg, Amir Zamir

Learning deep sensorimotor policies for vision-based autonomous drone racing **IROS 2023**

Jiawei Fu, Yunlong Song, Yan Wu, Fisher Yu, Davide Scaramuzza

Awards and Scholarships

Awards: Outstanding Graduate of Beijing ('20), Excellent Graduate of Tsinghua University ('20), Comprehensive Excellence Award ('17, '18, '19)

Scholarships: ETH Exchange Scholarship ('23), XCMG Scholarship ('19), Energy and Science Scholarship ('18), National Scholarship ('17)

Experience

Research Assistant, EPFL – Lausanne, Switzerland Mar. 2024 – May. 2024

- Implemented **co-design algorithms** of photoreceptors and control policy for indoor navigation in Habitat
- Deployed the control policy with an optimized mechanical design on a TurtleBot4 robot

Research Intern, Max Planck Institute for Informatics – Saarbrücken, Germany Apr. 2022 – Sept. 2022

- Built a **multi-sensor system** for **autonomous driving**, containing cameras, LiDARs, GPS, and event cameras
- Implemented a data collection pipeline including synchronization, calibration, and post-processing

Research Intern, University of Zurich – Zurich, Switzerland Sept. 2021 – Mar. 2022

- Developed a sensorimotor policy with **policy distillation** for autonomous drone racing
- Achieved robust racing performance against various visual disturbances with **contrastive representation learning**

Skills

Programming: Python, C++, PyTorch, Docker, Git, ROS, ROS2

Hardware: Mechanical Design, Circuit Design, Multi-device Synchronization